PRE RENAL vs RENAL AKI

Indices	Prerenal	Renal
Spot urine sodium (mEq/L)	<20	>40
Fractional excretion of sodium (FENa)	<1%	>2%
Fractional excretion of urea (FEUrea)	<35%	>35%
Urine osmolality (mOsm/kg)	>500	<350
BUN to serum creatinine ratio	>20:1	<10:1
Urine creatinine/plasma creatinine	>40:1	<20:1
Urinary sediment	Nil or hyaline casts only	Renal tubular epithelial casts, muddy brown granular casts

WHEN

ACUTE onset OLIGURIA + Nausea/Vomiting

+

NO FLANK PAIN / GROIN PAIN / DYSURIA

Kidney disease → AKI → Prerenal / Renal / Post-renal

Kidney disease → ~~Acute exacerbation of CKD~~

KFT → Urea - 130mg/dl
Creat → 2.1mg/dl

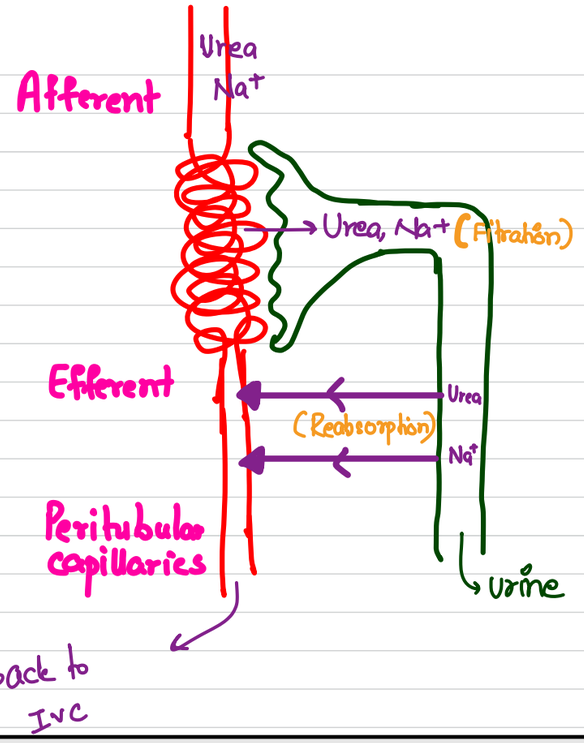
ACUTE vs CHRONIC

USG KUB

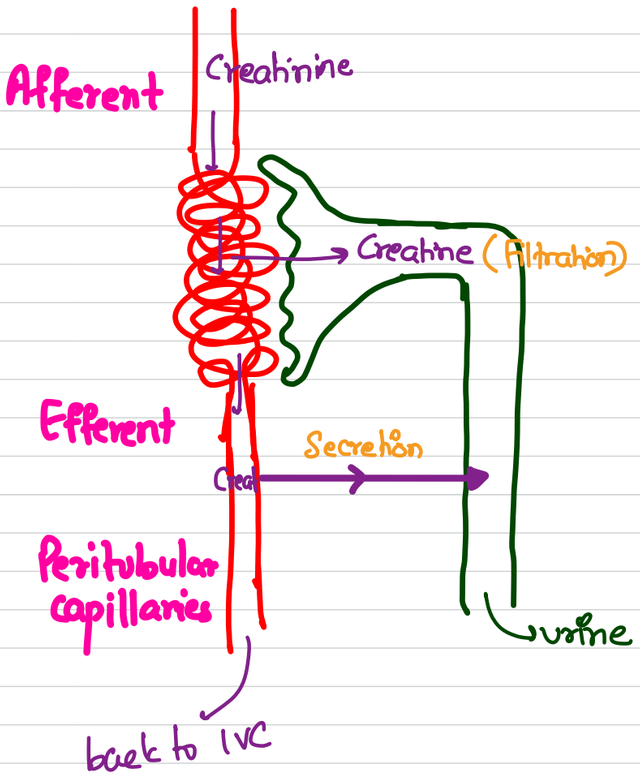
- N size & N CMD → r/o CKD ✓
- N urinary tract → r/o Post-Renal AKI ✓

PRE-RENAL vs RENAL

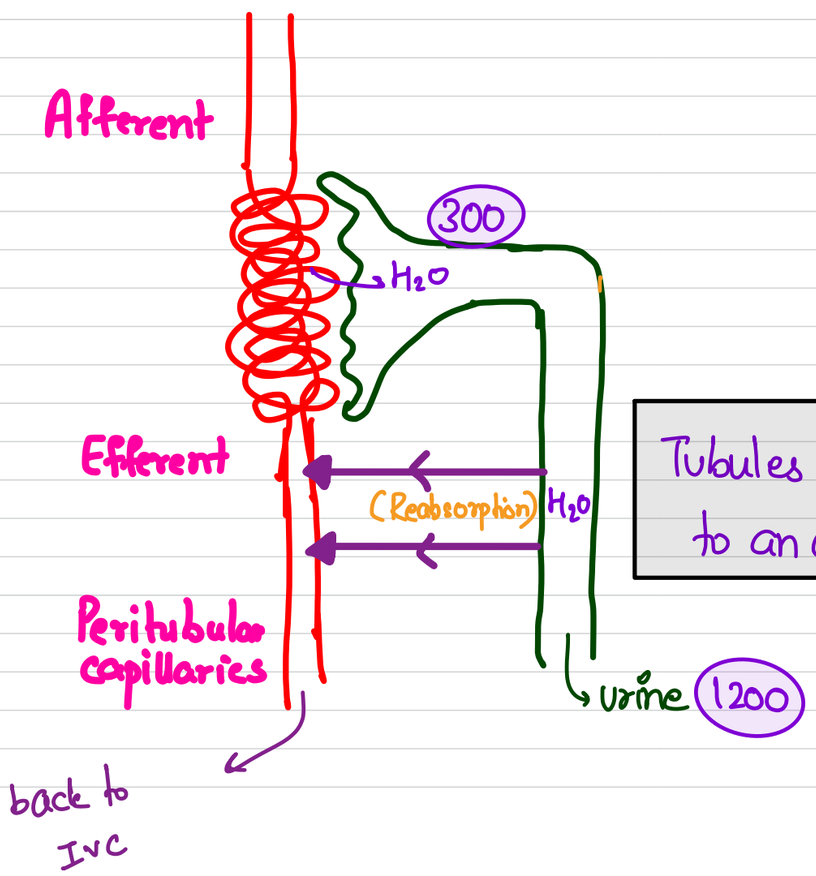
BASIC PHYSIOLOGY

① 

Urea & Na+ are reabsorbed into interstitium

② 

Creatinine is secreted from peritubular capillaries into the lumen

③ 

Tubules concentrate the urine from an initial osmolality of 300 in PCT to an osmolality of 1200 in urine

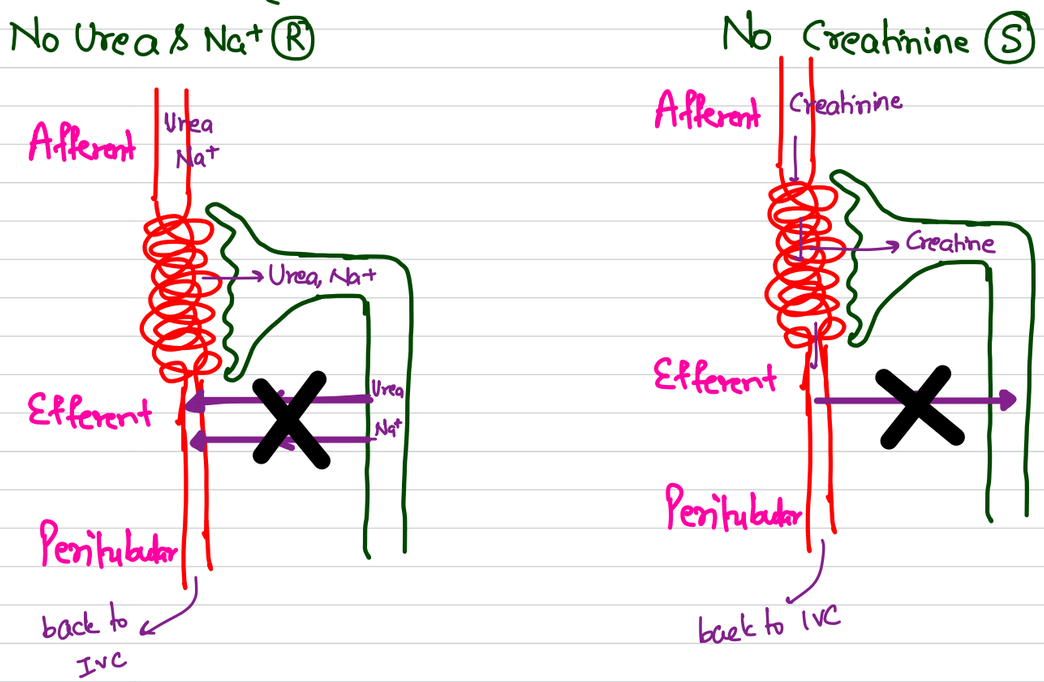
Renal - AKI

↓

MCC = Acute Tubular Necrosis (MC site → PCT)

↓

Tubules function lost



↑ Urea & Na^+ excreted in urine → Urine BUN ↑, Urine Na^+
↓ Urea & Na^+ (R) ⟶ S. BUN ↓, S. Na^+ ↓

↑ Creatinine goes back to IVC → U. Creat ↓
S. Creat ↑

① $\frac{\text{S. BUN} \downarrow}{\text{S. Creat} \uparrow}$ → <20

② U. Na^+ ↑ → >40

③ $\frac{\text{U. } Na^+ \uparrow}{\text{S. } Na^+ \downarrow}$ → ↑

⑤ $\frac{\text{U. Cr} \downarrow}{\text{S. Cr} \uparrow}$ → <20

④ $FeNa = \frac{\text{U } Na^+ \uparrow}{\text{Filtered } Na^+}$ → > 2%

⑥ Urine osmolarity → low (< 350)
(Urine concentrating capacity lost as tubules damaged)

A 38 year old woman presents with nausea, vomitting, fatigue, oliguria for the past three days. There is no history of dysuria, flank or groin pain. Patient has a history of severe diarrhea preceding to the onset of these symptoms.
Patient is not a k/c/o Hypertension, Diabetes, Kidney Disease

KIDNEY
AKI — Acute
CKD

A Kidney Function Test and Urine Analysis was done that showed
Urea- 130mg/dl ↑
Creat- 2.1mg/dl ↑) → ↓ Kidney Disease

Urinary Na+- 10meq/L
Urine osmolality- 800mOsm/kg

A further USG KUB was done that showed normally sized Kidneys with CMD maintained.

CKD
UT (N)
Post renal AKI

Pre- or Renal.

Urea → 130 — BUN — 130/2.8
Cr → 2.1
UNa+ -10 U.O — 800

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10

800

130

2·8x21

→ Pre-renal AKI

Pre-renal AKI